
Calibrated Risk Envelopes for LLM Quantization, and the Limits of Predicting Compression Damage

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Abstract

1 Quantizing a large language model reduces serving cost and costs some accu-
2 racy. Determining how much normally requires re-running a full benchmark suite,
3 which is the expense quantization was meant to avoid. We ask whether the accu-
4 racy delta can be predicted instead. Using 817 published evaluations spanning 38
5 base checkpoints, 6 model families and 6 quantization schemes, we report three
6 results. First, a *negative* one: per-model prediction carries almost no signal. Un-
7 der leave-one-family-out validation the best predictor, a per-scheme mean, beats
8 a global mean by 0.03pp of MAE (95% checkpoint-bootstrap CI [0.01, 0.05]),
9 and both ridge regression and gradient boosting are *worse* than guessing the av-
10 erage, because evaluation noise accounts for at least a third of the variance on 8
11 of 11 benchmarks. The features we tested are card-level (bit-width, scheme, size,
12 benchmark identity, headroom); we did not test predictors that use checkpoint-
13 derived features such as weight statistics or activation-outlier metrics. Second, a
14 positive one: conditioning only on the quantization scheme yields per-scheme
15 conformal-style intervals with 90.8% two-sided empirical coverage (119/131,
16 95% CI [84.5, 95.2]) on unseen checkpoints. Under the family definition of §3
17 the strict set splits into 7 in-family checkpoints (DeepSeek-R1-Distill of Llama
18 or Qwen bases) at 90.7% and 12 out-of-family at 91.1%; the checkpoint-cluster
19 95% CI on the difference is [-9, +11]pp and does not resolve whether the two
20 are equivalent. Within out-of-family, Gemma-3 alone (the only sub-group with
21 more than 10 rows) covers at 88.9% (40/45), 1.1pp below nominal. Those check-
22 points cover 4 of the 6 schemes, and W4A16 supplies 80 of the 131 rows. Each
23 (scheme, size-band) cell returns one of three verdicts: trusted; insufficient evi-
24 dence, where the interval is still shown but the scheme is demoted and flagged;
25 or refused, where the interval is withheld. Currently 6 of 17 cells are insufficient
26 evidence; on this corpus the third state, as defined in §4, does not fire. Third, and
27 most consequential for practice: the published corpus is outcome-truncated. Every
28 configuration in it is one that someone chose to release. We rented an A100 and
29 measured deliberately-faulted configurations, observing losses to -39.5pp against
30 a published worst case of -8.86pp—measured, we stress, only on models of 1.5B
31 and below, and on configurations engineered to fail rather than typical ones. This
32 is an existence proof that far worse outcomes exist than the corpus contains; it
33 does not estimate how common they are. We argue the resulting artifact should be
34 described as a calibrated historical baseline rather than a predictor, and we state
35 precisely why its coverage guarantee does not transfer to a practitioner’s own un-
36 tuned recipe. Code, data and the provenance of every reported figure are provided
37 as supplementary material.

1 Introduction

Post-training quantization is now routine. A practitioner choosing between W4A16, W8A8 and FP8 wants one number: how much accuracy will this cost? In practice the only way to know is to run the evaluation.

This paper asks whether that number can be forecast from the public record. Model publishers release quantized checkpoints with benchmark tables comparing the compressed model against its dense parent. Aggregated, these are hundreds of paired measurements of exactly the quantity of interest (850 extracted in our harvest).

Our contributions:

1. **A negative result** (§5). Per-model prediction of the accuracy delta *from card-level features* does not work, and we show why: the target is dominated by evaluation noise. Checkpoint-derived features (weight statistics, activation-outlier metrics) were not tested. We report this as the primary finding rather than burying it.
2. **A calibrated envelope that does work** (§6), conditioned on scheme alone, with measured coverage on held-out checkpoints and a three-state verdict for each (scheme, size-band) cell: *trusted*; *insufficient evidence*, where the interval is still shown but the scheme is demoted and the cell flagged as one the tool cannot judge; or *refused*, where the interval is withheld. Currently 6 of 17 cells are insufficient evidence; on this corpus the third state, as defined in §4, does not fire (see §4 for why).
3. **An existence proof that the corpus omits severe outcomes** (§7), via GPU experiments on deliberately-faulted configurations—data the published record structurally cannot contain. It shows that losses far beyond anything published are reachable on small models. It does not estimate how large the bias is: our attempt at a correction (in the supplement) did not work well enough to ship.
4. **An audit** (§9) recording errors we found in our own work, including ones that changed published numbers and claims.

2 Related Work

Predicting evaluations. BenchPress Zeng and Papailiopoulos [2026] establishes the mechanism closest to ours: predict an eval result, estimate whether to trust it, wrap it in a conformal interval, and decide whether to skip the real run. Their score matrix over 84 frontier models and 133 benchmarks is approximately rank-2. We do not claim novelty against them on mechanism. Their estimand is a model’s absolute score; ours is the paired delta between a model and its compressed self, and quantized checkpoints do not appear in their paper.

Conformal prediction on compressed models. Tong et al. [2026] apply conformal prediction directly to quantized and sparse LLMs, which is a closer domain than BenchPress. The estimand differs: they evaluate predictive uncertainty (coverage and set size) *after* running the compressed model. Ours is constructed over historical accuracy deltas to give an interval *before* running anything. Their Figure 2 plots the same quantity we predict, $\text{Acc}_{\text{compressed}} - \text{Acc}_{\text{dense}}$, but measures it. Their finding that compression decouples accuracy from uncertainty bounds what an accuracy-only estimate like ours can claim (§8).

Quantization methods. The checkpoints we harvest are produced with GPTQ Frantar et al. [2022] and SmoothQuant Xiao et al. [2022] via llm-compressor vLLM Project [2024]. Kumar et al. [2024] fit a functional form that predicts degradation from training and inference in varied precisions, using pretraining runs of models up to 1.7B parameters. We fit no such law: our envelope is empirical, per scheme, over published post-training-quantized checkpoints.

The source of our data. Kurtić et al. [2024] report over 500,000 evaluations of quantized Llama-3.1 models, finding recovery above 99% of the average OpenLLM v1 score for every scheme tested. As far as we can tell, that work does not address whether the published set of quantized models is a biased sample. Characterising that gap is the contribution of §7.

87 **Selection bias.** Survivorship bias has been identified and measured in a widely used retrieval
88 benchmark Gupta and MacAvaney [2022]. Conformal prediction’s guarantee is known to fail un-
89 der selection: conditional on a selection event, calibration data are no longer exchangeable with a
90 test point Barber et al. [2023], Jin and Ren [2024]. We apply this directly to our own artifact in §8.

91 3 Data

92 We harvest published model cards into rows of (model, scheme, benchmark, $\text{acc}_{\text{before}}$, $\text{acc}_{\text{after}}$), yield-
93 ing 850 rows, of which 817 remain after dropping 33 whose baseline accuracy is below 20% (near
94 chance, where a delta is uninformative), across 38 base checkpoints, 6 families and 6 schemes.

95 **An integrity gate.** Cards typically print a *Recovery %* column alongside the before/after pair.
96 This is redundant information, and we exploit it: for every three-column row we check $\text{recovery} \approx$
97 $100 \times \text{after/before}$ to a tolerance derived from the card’s own printed precision, and reject rows that
98 disagree. This caught spurious values that our parser produced under column-order variation be-
99 tween card templates, and internally inconsistent cards. Rejections are recorded rather than silently
100 dropped.

101 **A structural limitation, stated at the outset.** All rows come from a single publisher using a
102 single toolchain. Holding out a model family tests transfer across *models*; nothing in this design
103 tests transfer across *quantization practitioners*. §7 gives direct evidence that this distinction matters.

104 **What we mean by family.** Throughout, two checkpoints share a family when one is derived from
105 the other’s pretraining weights (via distillation, continued pretraining, quantization, or fine-tuning),
106 or when they share a common pretraining base checkpoint. A distillation is a member of its teacher’s
107 family; a new pretraining run is a new family, regardless of shared name. This is the mechanistic
108 definition: quantization damage depends on the weight distribution and activation-outlier structure
109 the checkpoint inherits, and derivative products inherit them, while new pretraining runs produce
110 different weight distributions and different quantization behaviour. In particular, under this def-
111 inition Gemma-3 and Gemma-2 are separate families (independent pretraining runs), Llama-3.1,
112 Llama-3.2 and Llama-3.3 all belong to the same Llama-3 family (Llama-3.2 1B/3B are pretraining-
113 time distillations from Llama-3.1 8B/70B logits; Llama-3.2 11B/90B are Llama-3.1 plus a vision
114 adapter; Llama-3.3-70B is a Llama-3.1-70B fine-tune – Meta’s own model cards state each of these
115 lineages), and DeepSeek-R1-Distill-Llama-8B is a member of that Llama-3 family (it uses Llama-
116 3.1-8B weights as initialization). The 6 training families are Gemma-2, Granite, Llama-3, Mistral,
117 Qwen-2.5 and Qwen-3. An earlier version of this paper counted Llama-3.1, Llama-3.2 and Llama-
118 3.3 as three separate families, so the corpus was reported as having 8 training families rather than 6,
119 and leave-one-family-out was run over 8 folds rather than 6; the audit note in §9 records the change
120 and its effect on the aggregate LOFO figure.

121 **The pathway is only sparsely documented.** A live scan on 2026-09-26 across the 439 RedHatAI
122 quantized-model cards currently published on Hugging Face (of which 102 contribute rows to our
123 harvested dataset) shows how sparsely the pathway that produced these outcomes is documented:
124 12 of 439 cards pin a quantization-library version, 33 mention activation reordering, 17 mention
125 dampening, and 198 of 439 report none of `num_calibration_samples`, `max_sequence_length`,
126 `activation_reordering` or `dampening`. On the 102-card subset that our dataset draws rows from, 0
127 pin a library version and 39 of 102 report none of those four parameters. (This RedHatAI-only
128 scan is distinct from the 10-publisher, 260-card publisher census reported in §9, which is an earlier
129 stable snapshot spanning multiple publishers and asks a different question.) The published record we
130 calibrate against is therefore a record of outcomes with the pathway that produced them substantially
131 undocumented on the publisher whose cards contribute the largest single share of our corpus, which
132 is one of the reasons an outcome-truncation argument (§7) cannot be closed from the corpus alone.

133 4 Method

134 **Conformal-style intervals.** We use the residual-quantile construction of split conformal re-
135 gression Lei et al. [2018]: given residuals $\{s_i\}_{i=1}^n$ and level α , the interval half-width is the

136 $[(n + 1)(1 - \alpha)]$ -th smallest score, which under a proper calibration/test split gives exact finite-
 137 sample coverage under exchangeability. Where n is too small for that index to exist, we return
 138 ∞ rather than a misleadingly finite band. The shipped per-scheme interval fits its centre and its
 139 half-width on the *same* rows (an in-sample residual-quantile band), so the split-conformal guaran-
 140 tee holds only for the explicit calibration-set variant used inside the leave-one-family-out audit; the
 141 empirical prospective figure below is what carries the shipped intervals.

142 **Mondrian (group-conditional) calibration.** Calibrating separately within categories is the Mon-
 143 drian construction of Vovk et al. [2005], applied to regression by Boström and Johansson [2020]. A
 144 single marginal interval advertised at 90% covers only 62% of NVFP4 rows under leave-one-family-
 145 out (recomputed live)—the users who most need it. We therefore calibrate per scheme.

146 **A three-state verdict.** For each (scheme, size-band) cell we measure coverage directly, and the
 147 tool returns one of three verdicts. A cell is *trusted* when measured coverage is adequate and ad-
 148 equately supported. It is *refused*—INSUFFICIENT CALIBRATION, no interval—when coverage is
 149 measurably poor: the checkpoint-bootstrap interval on its coverage lies wholly below the 85% line,
 150 on at least 3 checkpoints. It is *insufficient evidence* when the cell rests on fewer than 3 distinct
 151 checkpoints, or when a checkpoint-level cluster bootstrap 90% interval on its coverage straddles the
 152 85% line. In that case the evidence cannot support either of the other two verdicts. The tool still
 153 prints the all-sizes interval for such a cell, but demotes the scheme to the lowest tier and states that
 154 the cell cannot be judged; only a refused cell has its interval withheld.

155 Coverage is scored *one-sided*: how often the true delta stayed at or above the interval’s lower bound.
 156 The risk this tool bounds is accuracy loss, and a model that beats its envelope has exposed no one to
 157 anything, so counting that as a failure would flag cells that are merely outperforming. The interval
 158 is symmetric, so a 90% two-sided nominal corresponds to a one-sided nominal near 95%; pooled
 159 one-sided coverage across all 817 rows, with each row scored under 5 different calibration families,
 160 is 95.8%—essentially at that target, not above it.

161 Currently 0 of 17 cells are refused. Two cells were flagged under the earlier two-state rule, and nei-
 162 ther is a confident refusal under the three-state one. `w4a16|<2B` had one-sided coverage of 76.4%
 163 over 11 rows (two-sided 76.4%), but those rows come from only 2 checkpoints. `w8a16|>10B` had
 164 81.3% over 31 rows from 5 checkpoints (two-sided 73.5%), and its bootstrap interval, [47, 100]%,
 165 straddles the threshold. Both are now *insufficient evidence*. We had scoped the change to those
 166 two cells; applying the rule to every cell moved 6 of 17 into that state, including several whose
 167 point-estimate coverage looked comfortable. The same measurement and the same rule judge each
 168 scheme’s all-sizes coverage: 1 of the 6 schemes is insufficient evidence at that level and 0 is refused.
 169 The one-sided scoring and the three-state rule were proposed by an external commenter who identi-
 170 fies as an AI account collaborating with the OpenLanguageModel maintainers; we recomputed each
 171 of their figures from the raw rows before adopting either change, and one of their descriptions did
 172 not survive that check.

173 **Post-hoc thresholds, honestly labeled.** The three-state rule was proposed for two specific cells
 174 and then applied to every cell; the particular thresholds—85% for the refusal line, 3 checkpoints
 175 for the support floor, a 90% cluster bootstrap for the straddle test—were not derived from a first-
 176 principles argument, they were the values under which those two cells came out the way we thought
 177 they should. On this corpus one further consequence of that choice is that the *refused* state, as
 178 defined, does not fire: every cell with enough checkpoints to reach the second gate has a boot-
 179 strap upper bound comfortably above 85%, and the one cell whose upper bound does sit below it
 180 (`w4a16|<2B`, upper bound 83.3%) has too few checkpoints to reach that gate and lands in *insuffi-*
 181 *cient evidence* instead. So the current three states are, on this corpus, effectively two: *trusted* and
 182 *insufficient evidence*. This is a design finding, not a runtime bug, and we state it here rather than let
 183 a reviewer discover it.

184 **A support floor.** Coverage measured on many benchmarks from one checkpoint is not indepen-
 185 dent evidence; it is one quantization run viewed many ways. Cells resting on fewer than 3 distinct
 186 checkpoints cannot reach the top tier regardless of measured coverage. 4 cells are affected.

5 The negative result

Under leave-one-family-out validation, row-weighted MAE:

predictor	MAE (pp)	beats global mean?
per-scheme mean (shipped)	0.7227	yes
global mean (baseline)	0.7545	—
per-(scheme $\times$ benchmark)	0.7607	no
ridge regression	0.7639	no
per-benchmark mean	0.7647	no
gradient boosting	0.7878	no

The per-scheme mean’s advantage is small: 0.03pp, with a checkpoint-bootstrap 95% CI of $[0.01, 0.05]$ pp, and it is positive in 5 of the 6 held-out families. It is a distinguishable but modest gain. Ridge regression ($\alpha = 3.0$, standardised features, no cross-validated tuning) and histogram gradient boosting (`max_depth=3`, `max_iter=300`, `learning_rate=0.05`, `min_samples_leaf=15`, `l2_regularization=1.0`, `random_state=0`, no cross-validated tuning) are both *worse* than predicting the global average. Adding model size, bit-width, method, benchmark identity and headroom each degraded out-of-family performance. All features tested are card-level; we did not compute any checkpoint-level statistic (weight kurtosis, activation outliers, singular-value structure) and feed it in as a predictor. Those directions are open.

Why. Estimating per-benchmark evaluation noise from near-lossless schemes—whose true delta should be approximately zero—gives a noise floor that accounts for anywhere from 15% of the observed variance to all of it across 11 benchmarks. On GSM8K the deltas of supposedly lossless schemes have a standard deviation of 2.1pp, while the quantity being predicted has mean -0.39 pp. The target is a difference of two noisy quantities that largely cancel: the numerically smallest and hardest-to-predict summary of quantization damage available. Two cautions on the noise figures. They assume near-lossless schemes have a true delta near zero, so they are upper bounds on noise. And 5 of the 11 estimates rest on 10 or fewer rows; one, at 124%, exceeds the total variance, which we attribute to sampling error rather than treat as a finding.

Implication. From card-level features, the entire fitted artifact is 18 numbers. A raw per-scheme historical quantile band, with no model and no conformal machinery, is a serious competitor: it looked better covered in-sample, though under the strict protocol that advantage disappears (§9). What conformal contributes on the calibration-set variant used inside the leave-one-family-out audit is a finite-sample guarantee under exchangeability; the shipped intervals fit their centre and half-width on the same rows and do not carry that guarantee (§4). What the machinery contributes to the shipped artifact is a principled rule for small- n schemes, not accuracy. We therefore describe the result as a *calibrated historical baseline*, not a predictor.

6 Results

What this tool is for. The scenario it answers is: a practitioner is considering a scheme on a checkpoint from a family already represented in the published record and wants a calibrated range for the accuracy delta with an honest verdict on whether the record supports that range for that cell. Within that scope, each cell returns an interval and one of three tier labels; outside it, the tool declines to speak. It is not a scheme-safety certificate, does not bound a scheme’s worst case, and does not extrapolate to recipes or families the corpus has not already seen – §7 and §8 state each of those out-of-scope cases directly.

scheme	90% interval (pp)	worst observed	% losing >3pp	evals
fp8	[-1.11, +0.48]	-1.90pp	0.0%	80
w8a8_int	[-1.71, +1.07]	-3.32pp	1.0%	194
fp8_dynamic	[-1.44, +1.34]	-8.72pp	1.0%	200
w8a16	[-0.98, +0.81]	-3.04pp	1.2%	86
w4a16	[-2.87, +1.40]	-8.86pp	5.7%	193
nvfp4	[-4.10, +1.87]	-8.12pp	15.6%	64

Prospective validation. After freezing the method we harvested new checkpoints not used during development. The strict protocol (`verify/independent_check.py`) excludes a prospective card on one of two grounds: any card whose family label overlaps a training family (Llama-3.* as one family under Definition B, and Qwen-3), or any card whose format informed a parser fix (Llama-4). The two grounds are separate; only Llama-4 is excluded for the parser reason. Under this protocol two-sided coverage is $119/131 = 90.8\%$, 95% Clopper–Pearson CI $[84.5, 95.2]$, against a nominal 90% two-sided level. Note the layering: this filter runs by name-pattern regex; it aligns with the mechanistic family definition of §3 on the training-family names, and 7 of the 19 strict checkpoints (all DeepSeek-R1-Distill of a Llama-3.* or Qwen-2.5 base) are in-family derivatives under §3 that the name-pattern filter still lets through. The pooled figure therefore mixes “passed the strict filter” with “is in-family under §3”. The family-split reporting below applies §3 to what survives the filter and states both sides separately. The 131 strict rows come from 19 distinct checkpoints, which we hand-classify against the training-family list. Under the definition of §3: 7 in-family checkpoints (all DeepSeek-R1-Distill of a Llama-3.* or Qwen-2.5 base in training) contribute 75 rows at 90.7% two-sided coverage; 12 out-of-family checkpoints (Gemma-3, SmoLLM and SmoLLM3, NVIDIA-Nemotron-Nano-9B) contribute 56 rows at 91.1%. Both point estimates land near the nominal 90% two-sided level, but the checkpoint-cluster 95% CI on the in-minus-out difference is $[-9, +11]$ pp: at this sample size we cannot resolve whether in-family and out-of-family coverage are equivalent, and the paper does not claim they are. The aggregate out-of-family figure carries the same Simpson caveat as the pooled headline: Gemma-3 alone (the only sub-group with more than 10 rows) covers at 88.9% (40/45 rows from 8 checkpoints), 1.1pp below nominal, and SmoLLM/SmoLLM3 and Nemotron-Nano contribute 11 rows at 100% from 4 small-sample checkpoints, which pull the out-of-family aggregate up by 2.2pp above Gemma-3 alone. The classification is enumerated for these 19 checkpoints, not a general algorithm (§9). A from-scratch, standard-library-only reimplementation sharing no code with the pipeline reproduces this figure with zero disagreements. That checks the computation, not the validity of the design.

Merged-fold disclosure: Llama-3.1 vs 3.2 vs 3.3. Under Definition B the six-fold LOFO merges Llama-3.1, Llama-3.2 and Llama-3.3 into one Llama-3 fold. The merged mean can conceal that Llama-3.3 alone consistently sits several points below its cousins under every calibration family, so we break out the held-out Llama-3 fold by subgroup. Rows are the calibration family; columns are the three subgroups the merged fold contains.

calibration	Llama-3.1	Llama-3.2	Llama-3.3	merged
gemma-2	87.6%	87.5%	76.6%	85.7%
granite	95.2%	95.0%	85.1%	93.4%
mistral	96.2%	100.0%	91.5%	96.0%
qwen2.5	88.7%	85.0%	83.0%	87.2%
qwen3	100.0%	100.0%	93.6%	98.9%

Under every calibration family Llama-3.3 sits below both Llama-3.1 and Llama-3.2, and the merged mean is pulled up by the two better-covered subgroups. Under an 8-family reporting this weakness would show as Llama-3.3’s own held-out fold at 85.1% (the worst held-out family). Under Definition B it is folded into 92.2% on the merged Llama-3 mean; we report the breakdown here so the merge does not hide it.

What the prospective set does and does not cover. The 131 strict rows come from 19 quantized checkpoints, not 131 independent observations: a bootstrap that resamples checkpoints gives a 95% interval of $[85.7, 94.7]\%$, close to the row-level one. The rows cover only 4 of the 6 schemes; FP8 and NVFP4 have no prospective rows, so the figure says nothing about them.

	scheme	strict rows	checkpoints	covered (%)
266	w4a16	80	11	88.8
	w8a8_int	36	5	91.7
	fp8_dynamic	11	2	100.0
	w8a16	4	1	100.0

267 W4A16, the most common scheme, dominates the pooled figure and sits below the nominal 90%;
268 the other schemes rest on few checkpoints. Over all 186 harvested prospective rows, before the strict
269 filter, coverage is 90.9%, so the filter did not manufacture the headline. That said, the pooled 90.8%
270 is pulled up by 11 FP8-dynamic rows from 2 checkpoints and 4 W8A16 rows from 1 checkpoint,
271 both at 100.0% coverage: W4A16, at 80 of the 131 strict rows (61%), comes in at 88.8%, which is
272 below the nominal 90%. The pooled number should be read as an average across schemes with very
273 different sample sizes, not as a coverage claim uniform across schemes.

274 **What the envelope cannot do.** 0 of 6 intervals exclude zero: the tool can never state that a scheme
275 *will* cost accuracy. And 19 of 25 observed losses worse than 3pp fell *below* their interval floor—when
276 real damage occurs, the interval usually misses it. This is the single most important caveat in the
277 paper. The obvious mitigation, an asymmetric band, was tried and lost on both coverage and width
278 where it was finite (§9), so there is no drop-in fix on offer.

279 7 The measured left tail

280 The corpus contains only configurations someone chose to publish. To measure what it structurally
281 cannot show, we rented an A100 and quantized real models under deliberately-faulted recipes.

282 **Result.** Worst measured loss -39.5pp , against -8.86pp as the worst loss anywhere in the pub-
283 lished corpus. 16 of 36 faulted rows lost 3pp or more; separately, 2 of 9 rows from our correctly-
284 configured control arm did too. The dominant single effect was wrong scale granularity—per-tensor
285 instead of per-group.

286 **Scope, stated plainly.** This tail was measured only on models of 1.5B and below. It establishes
287 that catastrophic damage is reachable; it is not a measured bound for larger models.

288 **A confound we found and did not resolve.** Re-running MMLU under the published protocol
289 (5-shot, letter-scored) recovers accuracies in the range published on the cards: our Qwen2.5-0.5B-
290 Instruct scores 47.20 against the Instruct card’s 46.83 (the base-model card gives 47.57), and our
291 Qwen2.5-1.5B-Instruct scores 59.57. The harness question closed. Our control run then compared
292 Instruct-model checkpoints against Red Hat’s *base-model* W4A16 publications, because no Instruct
293 W4A16 checkpoints are published: our correctly-configured Instruct control is $2.6\times$ more damag-
294 ing than the base-model published W4A16 (mean gap -3.78pp , $z = 2.9$ and 5.3). This is not like-
295 for-like—Instruct models could plausibly be more fragile under 4-bit quantization than their base
296 counterparts—so the ratio is indicative rather than a clean estimate. A matched Instruct-vs-Instruct
297 control is deferred future work. The pattern we observed is consistent with approximately $171\times$ less
298 calibration data than production defaults, though the attribution is not independently tested. Conse-
299 quently, excess-over-control measures *bad recipe versus mediocre recipe*, not versus production. It
300 is an indicative anchor, not a bound, and we decline to publish a corrected point estimate: the cor-
301 rection moves by 5.70pp depending on the anchor set, and we have only 6 rows from two models.

302 8 Limitations

303 **The guarantee is conditional, and absent where it matters most.** Split conformal gives exact
304 coverage under exchangeability between calibration and test. Our calibration set is the set of config-
305 urations a publisher chose to release—a selection event. Conditional on selection, calibration points
306 are not exchangeable with an arbitrary test point and the guarantee does not transfer Barber et al.
307 [2023], Jin and Ren [2024]. It holds for a new checkpoint drawn from the same publication process.
308 It carries **no guarantee** for a recipe the reader tuned themselves and that no publisher would have
309 released—which is the case a practitioner most wants covered.

310 **Accuracy is not the whole of deployment risk.** Tong et al. [2026] show compression frequently
311 decouples accuracy from uncertainty: a model can hold accuracy while its prediction sets inflate. A
312 scheme this work calls low-risk on accuracy may still degrade uncertainty quality. We measure none
313 of that.

314 **One publisher, one toolchain, and why.** See §3. Our own control arm (§7) is direct evidence
315 that practitioner and calibration volume can dominate scheme choice. We measured how confining
316 this is: surveying 10 publishers and 260 quantized model cards, 10 of 222 cards from the other 9
317 publishers carry paired before/after evaluations, 9 of them Intel’s, so others do publish them. What
318 the other publishers almost never provide is a *Recovery* column, which is what lets each row be
319 checked against $100 \times$ after/before and rejected on disagreement: of their rows, 6 pass that gate,
320 all from 1 card, against 147 from the control publisher. The corpus is single-publisher not because
321 others publish nothing, but because the control publisher’s cards are, by a wide margin, the ones
322 published in a self-verifying form.

323 **What can and cannot be re-run.** The supplement recomputes every tagged figure from the data
324 files it ships. It cannot re-run the faulted-configuration experiments (they need a GPU; the recorded
325 results are included), the harvest from raw model cards (the cards are not redistributed), or the
326 prospective-card checks (those files are not included), and re-running the low-rank comparison men-
327 tioned in the supplement needs a checkout of BenchPress’s released code.

328 **An attempt at re-running the calibration-data test.** We attempted to reproduce Red Hat’s pub-
329 lished W4A16 checkpoints for two of the corpus’s most-measured base models, Qwen2.5-0.5B
330 and Qwen2.5-1.5B, using the `llm-compressor` recipe at its documented defaults, in order to test
331 whether the roughly $171 \times$ calibration-data ratio between our own control and production defaults
332 accounts for the observed excess damage. Neither of these two cards pins a library version or names
333 a calibration dataset; both record only the resulting weight representation (symmetric per-group
334 INT4, group size 128 on 1.5B and group size 64 on 0.5B). Neither attempt succeeded. A first run
335 at the current `llm-compressor` + `compressed-tensors` + `transformers` stack produced quan-
336 tized models scoring at or below four-way random chance on 5-shot MMLU (23–28% absolute),
337 well past the worst outcome anywhere in the corpus or the control arm, which we treat as a stack-
338 level anomaly rather than a calibration-data finding. A second run at an era-appropriate pin inferred
339 from the card wording (`llmcompressor==0.4.1`, `transformers==4.44.2`) could not resolve to a
340 working import: the pinned `llmcompressor` requires a class that was only added to `transformers`
341 in a later release, so the pin does not exist as a working combination. The calibration-data hypoth-
342 esis therefore remains what the Results section already calls it, consistent with the observation but
343 not independently tested. That the recipe recorded on these two cards is not sufficient to reproduce
344 even the working-stack question, let alone the answer, is directly consistent with the sparsity of
345 documentation reported in §3.

346 **Rows are not independent.** Up to sixteen benchmark rows come from a single quantization run.
347 We mitigate with group-conditional calibration and the support floor, and note that a bootstrap over
348 the 19 quantized checkpoints gives [85.7, 94.7]%, close to the row-level interval, but the unit of anal-
349 ysis remains a row. The leave-family-out coverage measurement has the same dependence: each
350 row is scored once under each of 5 calibration families, so the 4085 evaluations are not indepen-
351 dent, and every coverage judgment in this paper resamples whole checkpoints rather than counting
352 evaluations.

353 9 Audit

354 We record errors found in our own work, including those that changed published claims. This
355 section exists because the alternative—reporting only what survived—would misrepresent how the
356 result was obtained.

357 Errors that changed a published number.

- 358 • *The training corpus originally counted Llama-3.1, Llama-3.2 and Llama-3.3 as three sep-*
359 *arate families.* Meta’s own model cards state that Llama-3.2 1B/3B were pretraining-time

distillations from Llama-3.1 8B/70B logits, that Llama-3.2 11B/90B are Llama-3.1 plus a vision adapter, and that Llama-3.3-70B is a Llama-3.1-70B fine-tune. Under the mechanistic definition adopted in §3 (a distillation is a member of its teacher’s family; a fine-tune inherits the base’s family) they therefore belong to a single Llama-3 family, and the corpus has 6 training families rather than 8. Leave-one-family-out is now run over 6 folds rather than 8. The prospective coverage figures are unchanged (the strict subset never contained a Llama-3.2 or Llama-3.3 checkpoint), but every LOFO-derived figure moves: the aggregate LOFO coverage across held-out families changes from 87.3% to 88.2%, and the previously-worst held-out fold Llama-3.3 (85.1%) folds into the merged Llama-3 fold at 92.2%. Per-family LOFO on the merged fold is reported so this weakness does not disappear from view.

- *A retracted headline, then re-partitioned.* An early claim that all 19 held-out checkpoints came from families never used in construction was wrong. Under the definition adopted in §3, 7 of the 19 are derivatives of a training family and 12 are new pretraining runs; the point estimates on both sides land near nominal but the difference between them is not resolvable at this sample size (§6).
- *Independent implementations shared a float comparison.* The prospective-coverage figure sits on a floating point boundary at one row: gemma-3-1b-it W4A16 on TruthfulQA has an observed delta of exactly +1.40pp and a scheme upper bound of +1.40pp; in real arithmetic that is a closed-interval hit, but in float arithmetic the computed upper bound is +1.3999999999999997, so the comparison delta $\leq$ hi returned False. All three reimplementations (the pipeline, the standard-library re-check, and the audit-side recomputation) agreed on the same 118/131 because they shared the strict comparison; the agreement was consistency, not correctness. The comparison sites were patched to add a 10^{-9} tolerance on each bound, and the strict figure is now 119/131. This is a real limit of independent reimplementations as a check: when two implementations share an assumption, they will agree on results the assumption is wrong about.
- *The family classifier is enumerated.* The in-family / out-of-family split of the 19 strict prospective checkpoints was decided by inspecting each card and matching against the training-family list; the code that emits the split is a regex specific to this set (DeepSeek-R1-Distill-(Llama|Qwen)). It is not a general implementation of the §3 definition. A prospective checkpoint outside this pattern (e.g. a distillation from another base, a fine-tune with an idiosyncratic name) would need a manual re-classification. The definition itself is fixed; the code path applies it to this set only.
- *Two parser bugs.* A benchmark label typo cost 11 training rows; separately, MATH-500 was matched as Math-Lvl-5, contaminating 7 test rows with scores from a different benchmark.
- *A centring mismatch.* Calibration measured residuals about the stratum mean while intervals centred on the scheme mean, mis-sizing every calibrated width. Found only by independent reimplementations.
- *A column-order fault.* One card template orders columns [recovery, base, quant], which our parser turned into a spurious −68pp delta. The integrity gate rejected 10 of 11 rows on that card, passing only the one where base = quant made the check degenerate.
- *A retracted single-publisher claim.* We had asserted that one publisher was close to the only source of paired evaluations. A census of 10 publishers found 10 paired-evaluation cards outside the control publisher, 9 of them Intel’s; what differs is whose rows can be verified (§8).
- *A bootstrap on the wrong statistic.* The checkpoint-level bootstrap in our coverage audit re-sampled the two-sided indicator while the verdict was judged on the one-sided statistic. An external commenter asked for it to wrap the one-sided figure, which exposed the mismatch (§4).
- *Inflated pair counts.* An earlier version of the tool reported the number of (test row, calibration family) evaluations as if it were the number of rows: with 6 families each row is scored 5 times, so 4085 evaluations came from only 817 rows. Every row-count claim was recomputed on `distinct_rows`.

415 • *Two levels judged differently.* Cells were judged on the one-sided coverage statistic and
416 refused when the point estimate fell below the threshold, but schemes had been judged on
417 the two-sided statistic and refused only when the entire bootstrap interval fell below it. A
418 single classifier now judges both levels the same way, and the envelope’s refused list is a
419 computed copy of the ranking tool’s, checked by a test.

420 **A regression we nearly shipped.** An audit indicated that a raw asymmetric empirical band cov-
421 ered better than the shipped symmetric conformal interval (93.4% versus 89.6%). Since quantization
422 damage is left-skewed, a symmetric band is obviously the wrong shape, and we were about to switch.
423 On measuring first, the empirical band returns an *infinite* interval on 43% of rows, because a two-
424 sided empirical index requires more calibration points than are often available. An infinite interval
425 covers by construction. On rows where both are finite, the original wins on coverage *and* width. The
426 change was reversed.

427 **An overclaimed guarantee.** The artifact was described as carrying a coverage guarantee without
428 qualification. Checking the method against the conformal literature rather than against our own
429 earlier work showed the guarantee is conditional on exchangeability that selection breaks. Corrected
430 in §8.

431 **Provenance discipline.** Two parser fixes were informed by inspecting rows that were already in
432 the test set. Rather than silently retain them, we record which fixes were test-informed and exclude
433 the affected family from the strict headline.

434 **An attempted extension of the adversarial arm to 3B.** On a fresh 40 GB A100 with a fresh
435 dependency install, we re-ran the adversarial sweep to (i) verify that yesterday’s numbers repro-
436 duce and (ii) add a Qwen2.5-3B-Instruct row. The regression check passed: base evaluations
437 on Qwen2.5-0.5B-Instruct and SmolLM-135M-Instruct reproduced byte-for-byte, deterministic-
438 quantization recipes (per-tensor RTN, w8a8 without smoothing) reproduced byte-for-byte, and
439 stochastic GPTQ recipes stayed within $\leq 0.85\text{pp}$ of yesterday, consistent with fp16 Hessian-
440 accumulation noise. The 3B extension did not land: base evaluations on Qwen2.5-3B-Instruct
441 completed, but all five quantization recipes hit `torch.cuda.OutOfMemoryError` during the paired-
442 evaluation forward pass at batch size 2, not during quantization itself; the fp16 model weights
443 ($\approx 6\text{ GB}$), PyTorch’s caching allocator fragmentation from unloaded prior states, and activation
444 memory for a batch of 2 long sequences together exceeded the 40 GB budget with roughly 10 GB
445 of free memory needed for the next allocation. A single 80 GB card would be expected to accom-
446 modate this, so the constraint is hardware, not methodology. The adversarial arm therefore remains
447 bounded at $\leq 1.5\text{B}$ as before; §8 states this scope directly.

448 **Mechanical verification.** The figures this paper reports as results are macros generated from a
449 claims registry that recomputes each value from source artifacts, and the accompanying code fails
450 a test if an outward-facing document contains an untraced numeric literal. The exceptions are his-
451 torical figures in the error list above (row counts and the -68pp delta), which are quoted from the
452 provenance record rather than recomputed.

453 10 Conclusion

454 Per-model prediction of quantization accuracy damage from card-level features does not work on
455 published data, and the reason is largely measurement noise rather than model capacity; predictors
456 that use checkpoint-level statistics were not tested and remain open. What does work is narrower: a
457 per-scheme calibrated envelope that is validated at 90.8% two-sided coverage on unseen checkpoints
458 for 4 of the 6 schemes. In-family and out-of-family point estimates both land near nominal (90.7%
459 and 91.1% respectively), but the sample size does not distinguish them, and Gemma-3, the largest
460 out-of-family sub-group, covers at 88.9%, 1.1pp below nominal. The envelope also says which of
461 its own cells it cannot judge. Each cell is trusted; or insufficient evidence, in which case the interval
462 is still shown but flagged and the scheme demoted; or refused, in which case the interval is withheld.
463 Today 6 of 17 cells are insufficient evidence; on this corpus the refusal state does not fire (§4). The
464 more useful contribution may be the limitation itself—the published record of quantized models is
465 outcome-truncated, and we measured, on deliberately faulted recipes for models up to 1.5B, losses

466 more than four times worse than anything it contains. A risk estimate built on that record is a floor
467 on risk, not a ceiling, and should say so.

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